

UST-BARGUZIN, Russian Federation

WMO: 306350

Lat: 53.417N

Lon: 109.017E

Elev: 461

StdP: 95.91

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.1	(c) -30.5	(d) -35.4	(e) 0.1	(f) -33.0	(g) -33.0	(h) 0.2	(i) -30.3	(j) 6.7	(k) -14.5	(l) 5.9	(m) -16.7	(n) 1.2	(o) 90	(p) 0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.7	(c) 26.1	(d) 17.6	(e) 23.8	(f) 16.9	(g) 21.7	(h) 16.2	(i) 18.9	(j) 23.8	(k) 17.8	(l) 21.9	(m) 16.9	(n) 20.6	(o) 2.6	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.0	20.8	16.3	12.2	19.8	15.4	11.6	18.9	56.1	23.7	52.4	22.0	49.5	20.7	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.7	5.0	4.4	-36.9	31.0	3.5	1.9	-39.4	32.4	-41.5	33.5	-43.5	34.5	-46.0	35.9
			-36.6	20.7	3.0	1.2	-38.7	21.6	-40.5	22.3	-42.2	23.0	-44.4	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.1	-20.2	-18.4	-10.7	-0.8	5.3	12.0	16.3	15.2	8.8	1.0	-8.0	-15.3	(6)
(7)		DBStd	13.41	5.86	6.29	6.43	4.05	3.18	3.47	3.08	2.99	3.55	3.95	5.82	6.08	(7)
(8)		HDD10.0	4532	938	795	642	323	149	16	0	2	64	278	542	784	(8)
(9)		HDD18.3	7133	1196	1028	900	573	403	191	79	106	286	536	792	1042	(9)
(10)		CDD10.0	469	0	0	0	0	5	78	196	161	28	1	0	0	(10)
(11)		CDD18.3	27	0	0	0	0	0	2	17	8	0	0	0	0	(11)
(12)		CDH23.3	245	0	0	0	0	0	36	143	64	2	0	0	0	(12)
(13)		CDH26.7	46	0	0	0	0	0	4	32	10	0	0	0	0	(13)
(14)	Wind	WSAvg	2.0	2.1	1.7	1.7	1.9	2.0	1.8	1.8	1.7	2.0	2.0	2.5	2.7	(14)
(15)	Precipitation	PrecAvg	372	13	5	6	13	17	34	78	65	40	25	36	37	(15)
(16)		PrecMax	493	52	17	22	40	43	118	229	129	100	57	68	60	(16)
(17)		PrecMin	274	4	0	0	2	3	1	10	13	15	2	11	12	(17)
(18)		PrecStd	60	11	4	6	9	10	27	63	28	19	16	16	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.3	-0.8	6.2	13.1	20.0	27.1	30.0	27.9	22.2	13.2	6.4	1.5	(19)
(20)			MCWB	-4.2	-3.3	1.6	5.8	11.0	16.5	18.9	18.0	14.5	8.4	3.4	-0.9	(20)
(21)		2%	DB	-5.7	-4.0	3.0	9.3	15.9	23.9	26.9	25.0	19.1	10.7	4.4	-1.5	(21)
(22)			MCWB	-7.1	-5.6	-0.1	4.2	9.0	15.3	18.3	17.5	13.7	7.1	1.9	-3.4	(22)
(23)		5%	DB	-8.9	-6.3	0.9	7.1	13.8	21.2	24.6	22.6	16.8	8.8	2.1	-3.5	(23)
(24)			MCWB	-9.9	-7.6	-1.7	3.0	7.9	14.6	17.8	16.6	12.5	5.7	-0.1	-5.1	(24)
(25)		10%	DB	-11.8	-8.7	-1.1	5.3	11.8	18.8	22.4	20.7	15.0	7.2	0.2	-6.0	(25)
(26)			MCWB	-12.5	-9.7	-3.1	1.9	6.8	13.6	17.0	16.0	11.3	4.4	-1.7	-7.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.7	-3.1	2.1	6.7	11.6	18.1	20.9	20.3	16.1	9.5	3.8	-0.8	(27)
(28)			MCDWB	-2.5	-1.0	5.3	12.3	17.9	23.8	26.1	25.3	20.0	12.5	6.4	1.3	(28)
(29)		2%	WB	-6.9	-5.5	0.1	4.6	9.7	16.5	19.6	18.2	14.4	7.5	2.1	-3.4	(29)
(30)			MCDWB	-5.7	-4.1	3.0	8.6	14.7	21.6	24.7	22.6	18.2	10.0	4.2	-1.7	(30)
(31)		5%	WB	-9.9	-7.6	-1.5	3.2	8.4	15.2	18.5	17.4	13.1	6.1	0.0	-5.1	(31)
(32)			MCDWB	-8.9	-6.4	0.7	6.5	12.9	20.0	22.9	21.2	16.0	8.5	1.9	-3.5	(32)
(33)		10%	WB	-12.5	-9.7	-3.0	2.2	7.2	14.0	17.6	16.5	11.8	4.6	-1.6	-7.3	(33)
(34)			MCDWB	-11.8	-8.7	-1.1	5.0	11.2	18.2	21.4	20.0	14.6	7.0	0.2	-6.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	11.1	11.5	8.9	9.7	9.4	8.7	8.9	9.0	7.7	6.4	6.6	(35)
(36)			MCDBR	9.8	11.4	10.5	11.5	14.0	14.7	13.3	12.9	12.1	10.4	8.3	9.0	(36)
(37)			MCWBR	8.9	9.9	7.6	7.3	8.1	7.4	6.2	6.8	7.3	7.5	6.7	7.9	(37)
(38)		5% WB	MCDBR	9.8	11.2	10.1	10.6	12.4	12.0	10.8	10.7	10.3	9.8	8.2	9.0	(38)
(39)			MCWBR	8.9	9.9	7.5	7.1	7.7	6.9	5.7	6.2	6.6	7.5	6.9	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.264	0.294	0.322	0.371	0.431	0.448	0.430	0.391	0.343	0.312	0.281	0.261	(40)	
(41)		taud	2.076	2.056	2.085	2.183	2.113	2.174	2.282	2.370	2.453	2.396	2.140	2.035	(41)	
(42)		Ebn at Noon	687	785	848	855	817	806	815	830	834	780	658	597	(42)	
(43)		Edh at Noon	71	102	123	129	146	140	123	107	87	73	66	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.98	1.97	3.39	4.73	5.72	6.12	5.58	4.67	3.36	1.92	0.94	0.64	(44)	
(45)		RadStd	0.08	0.12	0.13	0.31	0.35	0.37	0.47	0.21	0.27	0.19	0.10	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.93	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	-1.82	-2.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page